**Shreesha G. Bhat**

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## Research Interests

Distributed Systems, Data Streaming, Systems for Agentic AI, Machine Learning Systems, Storage Systems, CXL

## Education

**University of Illinois Urbana-Champaign** 2023 - 2028 (expected)
*PhD in Computer Science | Advisors: Ram Alagappan, Aishwarya Ganesan | CGPA: 4.0/4.0*

**Indian Institute of Technology, Madras** 2018 - 2023
*Dual Degree (BTech + MTech) in Computer Science & Engineering | Advisor: Kartik Nagar | CGPA: 9.66/10.00*

## Publications

**SOSP '26**
AgileLog: A Forkable Shared Log for Agents on Data Streams
*Shreesha G. Bhat, Tony Hong, Michael Noguera, Aishwarya Ganesan, Ramnatthan Alagappan*
The 32nd Symposium on Operating Systems Principles (SOSP) 2026

**OSDI '26**
MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory
*Jiyu Hu, Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, Aishwarya Ganesan* (*equal contribution)
The 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI) 2026
**Jay Lepreau Best Paper Award Nominee**

**EuroSys '26**
A Logically Disaggregated Cache for Replicated Storage Systems
*Kiran Hombal, Henry Zhu, Shreesha G. Bhat, Neil Kaushikkar, Ramnatthan Alagappan, Aishwarya Ganesan*
The 21st European Conference on Computer Systems (EuroSys) 2026

**OSDI '25**
Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering
*Shreesha G. Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan*
The 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI) 2025
**Covered on the Metadata Blog**

**SOSP '24**
LazyLog: A New Shared Log Abstraction for Low-Latency Applications
*Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan* (*equal contribution)
The 30th Symposium on Operating Systems Principles (SOSP) 2024
**Best Paper Award | Invited to Transactions on Computer Systems |**
**Inspired Lightning Topics in Warpstream**

**FMCAD '23**
Automating Cutoff-based Verification of Distributed Protocols
*Shreesha G. Bhat, Kartik Nagar*
Formal Methods in Computer-Aided Design (FMCAD) 2023

**DISC '21**
Brief Announcement: Automating and Mechanising Cutoff Proofs for Parameterized Verification of Distributed Protocols
*Shreesha G. Bhat, Kartik Nagar*
35th International Symposium on Distributed Computing (DISC) 2021

## Workshop Publications

**CAIS '26 SAO**
Agents for Data Streaming Tasks: The Missing Pieces
*Shreesha G. Bhat*, Landon Johnson*, Michael Noguera*, Aishwarya Ganesan, Ramnatthan Alagappan* (*equal contribution)
Supporting our AI Overlords Workshop at CAIS 2026 (SAO) May 2026
**Spotlight**

## Refereed Journal Publications

**TOCS '25**
LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications
*Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan* (*equal contribution)
ACM Transactions on Computer Systems (TOCS) August 2025
**Invited for Publication**

## Research Experience

**Research @ Distributed and Storage Systems Lab | UIUC** *Aug '23 – present*
*Research Assistant | Guide: Ram Alagappan, Aishwarya Ganesan*

· Designed and implemented multiple shared log abstractions and systems for modern applications.
· Contributed to projects on CXL data sharing and replicated storage systems.

**Database Team** | **OpenAI** *May '26 - Aug '26*
*Member of Technical Staff (Intern)*

**Networking and Distributed Systems Lab** | **Hewlett Packard Labs** *May'25 – August'25*
*Research Associate Intern* | *Guide: Puneet Sharma*

· Designed distributed runtime infrastructure for LLM-based multi-agent systems, addressing agent coordination, fault recovery, and execution observability.

**Improving Cloud Reliability through Systematic Testing** | **Microsoft Research India** *Aug '22 – Jul '23*
*Research Intern* | *Guide: Akash Lal*

· Worked on improving reliability of Azure Cloud Services using concurrency testing tools such as Coyote for C++ programs.
· Built a deterministic concurrency testing framework for a production scale replication library, Azure RSL, which provides an implementation of the Paxos consensus algorithm. Implemented several optimizations to improve state-space coverage.

**Parameterized Verification of Distributed Protocols** | **IIT Madras** *Sep '20 – Jul '23*
*Young Research Fellow* | *Guide: Kartik Nagar*

· Investigated cutoff-based techniques for verifying that distributed protocols meet their specification irrespective of the size of the parameter they are instantiated with (such as number of nodes).
· Proposed a framework to mechanize simulation based proofs for cutoffs and applied the approach on a variety of distributed protocols using Z3 as a backend SMT solver.

## Awards & Achievements

· Awarded the **Best Paper Award** at SOSP 2024 for the paper titled *LazyLog: A New Shared Log Abstraction for Low-Latency Applications*.
· Selected as a undergraduate research fellow as part of the (YRF) program.
· Secured prizes for excellent academic performance in the 1st, 2nd, 7th and 8th semesters at IIT Madras.
· Secured All India Rank of **851** and **619** in JEE (Joint Entrance Examination) Advanced & Mains 2018.
· Qualified for KVPY fellowship with an All India Rank of **142**.
· Among the **top 300** in India qualified to write national olympiad examinations for Physics, Chemistry and Astronomy (INPhO, INChO, INAO).

## Service

· **Artifact Evaluation Committees:** OSDI'24, ATC'24, FAST'26 (Spring and Fall)
· **Shadow Program Committees:** EuroSys'26
· **Student Volunteer:** SOSP'24

## Teaching

· **CS598 Storage Systems, TA, Fall 2025:** Held office hours, graded assignments, and led lecture sessions.

## Invited Talks

· **New Shared Log Abstractions for Modern Latency-Critical Applications**
*MadSystems (Systems Reading Group), University of Wisconsin-Madison* *November 2, 2026*
*Location*: Online
· **AgileLog: A Forkable Shared Log for Agents on Data Streams**
*Confluent Current 2026* *November 2026*
*Location*: San Francisco, CA
*Links*: Session
*Systems Group, University of Texas at Austin* *October 12, 2026*
*Location*: Austin, TX
*Stream Processing with Apache Kafka, Samza, and Flink Meetup hosted by LinkedIn* *June 24, 2026*
*Location*: Sunnyvale, CA + online
*Links*: Meetup Event | YouTube

## Grants

· **Conference Grants:** OSDI'25, SOSP'24, SOSP'26